INFOSYS SE QUANT VAULT • PERMUTATIONS, COMBINATIONS & PROBABILITY

Counting Principle, Arrangements, Selections, Independent Events • Total Solved Questions: 32

[Campus Monk Day 01 Mock Test]

Topic: *Quantitative Aptitude*

Q1. In the given questions, two quantities are given, one as 'Quantity 1' and another as 'Quantity 2'. You have to determine relationship between two quantities and choose the appropriate option: Question: Two dices are rolled simultaneously. Quantity 1: Probability that the sum of the numbers that appeared is a multiple of 5. Quantity 2: 1/6

(A) Quantity 1 > Quantity 2
(B) Quantity 1 ≥ Quantity 2
(C) Quantity 2 > Quantity 1
(D) Quantity 2 ≥ Quantity 1

Correct Answer: Option (A) : Quantity 1 > Quantity 2

■ **Step-by-Step Logic:** Step 1: Determine the total sample space size n(S). Step 2: Calculate the favorable event outcomes n(E) using combinatorial counting principles (nCr or nPr). Step 3: Compute Probability P(E) = n(E) / n(S) or evaluate the distinct arrangements. Step 4: Simplifying the fraction matches Option (A) (Quantity 1 > Quantity 2).

[Campus Monk Day 02 Mock Test]

Topic: *Quantitative Aptitude*

Q2. In an envelope there are 5 green, 3 yellow and 4 pink tablets. 3 tablets are picked at random.Quantity I: The probability that 2 tablets are yellow in colour and 1 tablet is pink in colour.Quantity II: The probability that all the tablets are green in colour.

(A) ---
(B) - (C)
(D) - (E)

Correct Answer: Option (A) : ---

■ **Step-by-Step Logic:** Step 1: Determine the total sample space size n(S). Step 2: Calculate the favorable event outcomes n(E) using combinatorial counting principles (nCr or nPr). Step 3: Compute Probability P(E) = n(E) / n(S) or evaluate the distinct arrangements. Step 4: Simplifying the fraction matches Option (A) (---).

[Campus Monk Day 03 Mock Test]

Topic: *Quantitative Aptitude*

Q3. A bag contains 5 red and 7 white balls. Four balls are drawn out one by one and not replaced. What is the probability that they are alternatively of different colours?

(A) 7/99
(B) 11/99
(C) ---
(D) 19/99

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Step 1: Determine the total sample space size n(S). Step 2: Calculate the favorable event outcomes n(E) using combinatorial counting principles (nCr or nPr). Step 3: Compute Probability P(E) = n(E) / n(S) or evaluate the distinct arrangements. Step 4: Simplifying the fraction matches Option (C) (---).

[Campus Monk Day 06 Mock Test]

Topic: *Quantitative Aptitude*

Q4. A team of four players are to be selected from among four girls A,B,C,D and four boys P,Q,R,S such that there would be at least two girls in the team and the team selection should meet the following conditions i. B cannot go with R ii .D and P must go together iii S cannot be put up with A iv. R and C must be together

(A) QBSC
(B) DAPQ
(C) RCQS
(D) BPRA

Correct Answer: Option (B) : DAPQ

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (B) (DAPQ).

[Campus Monk Day 06 Mock Test]

Topic: *Quantitative Aptitude*

Q5. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there?

(A) 50
(B) 100
(C) 150
(D) 200

Correct Answer: Option (C) : 150

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (C) (150).

[Campus Monk Day 07 Mock Test]

Topic: *Quantitative Aptitude*

Q6. To solve a crime, police interrogate both Felix and Victor. Felix speaks the truth in 60% of cases, and Victor in 40% of cases. Police can catch the criminal only if at least one of them speaks the truth. What is the probability that the police catch the criminal?

(A) 13/25
(B) 21/25
(C) 6/25
(D) 19/25

Correct Answer: Option (D) : 19/25

■ **Step-by-Step Logic:** Step 1: Identify principal, rates, and percentage shifts. Step 2: Formulate the financial relationship using Profit = SP - CP or A = P(1 + r/100)^t. Step 3: Substitute the parameters and simplify the fractional/percentage relations. Step 4: The calculated outcome matches Option (D) (19/25).

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q7. Question : A basket contains 6 White 4 Black 2 Pink and 3 Green balls. If four balls are picked at random, what is the probability that at least one is Black?

(A) 69/91
(B) 80/91
(C) 21/91
(D) 55/91

Correct Answer: Option (A) : 69/91

■ **Step-by-Step Logic:** Step 1: Determine the total sample space size n(S). Step 2: Calculate the favorable event outcomes n(E) using combinatorial counting principles (nCr or nPr). Step 3: Compute Probability P(E) = n(E) / n(S) or evaluate the distinct arrangements. Step 4: Simplifying the fraction matches Option (A) (69/91).

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q8. Question : A bag contains Rs. 4100 in the form of Rs. 5, Rs. 2 and Rs. 1 coin. The number of coins is in ratio 4: 6: 9. So, find the number of 1 Rs. coins.

(A) 40
(B) 500
(C) 900
(D) None of these

Correct Answer: Option (C) : 900

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (C) (900).

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q9. Directions : Study the following information carefully to answer these questions. A, B, C, D, E, F, G and H are sitting around a circle, facing the centre. F sits to the immediate right of D and third to the left of A. G sits third to the left of D who does not sit next to E. B sits next to G but not next to D. C does not sit next to either D or A. Question : Starting from A's position, if all the eight are arranged in alphabetical order in clockwise direction, the seating position of which of the following (excluding A) would not change?

(A) B
(B) C
(C) D
(D) H

Correct Answer: Option (B) : C

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (B) (C).

[College ITM Mock Paper 1]

Topic: Mathematical Critical Thinking

Q10. In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?

(A) 5400
(B) 3600
(C) 10800
(D) 2700

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]

Topic: Mathematical Critical Thinking

Q11. A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?

(A) 90
(B) 100
(C) 105
(D) 120

Correct Answer: Option (A) : 90

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]

Topic: Mathematical Critical Thinking

Q12. Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?

(A) 4/11
(B) 1/3
(C) 5/11
(D) 4/9

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]

Topic: Mathematical Critical Thinking

Q13. A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?

(A) 3/11
(B) 1/4
(C) 2/11
(D) 5/22

Correct Answer: Option (A) : 3/11

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]

Topic: Mathematical Critical Thinking

Q14. In how many ways can the letters of the word MANAGEMENT be arranged so that all the vowels come together?

(A) 7560
(B) 15120
(C) 3780
(D) 30240

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]

Topic: Mathematical Critical Thinking

Q15. From a group of 6 men and 5 women, a team of 4 is to be selected such that it contains at most 2 women. In how many ways can this be done?

(A) 265
(B) 295
(C) 310
(D) 325

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]

Topic: Mathematical Critical Thinking

Q16. Three coins are tossed together, and it is known that at least one head appeared. What is the probability that exactly two heads appeared?

(A) 3/7
(B) 1/2
(C) 3/8
(D) 2/7

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]

Topic: Mathematical Critical Thinking

Q17. A box contains 6 defective and 14 non-defective bulbs. Two bulbs are drawn at random without replacement. What is the probability that at least one is defective?

(A) 99/190
(B) 91/190
(C) 33/95
(D) 29/95

Correct Answer: Option (A) : 99/190

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]

Topic: Mathematical Critical Thinking

Q18. In how many ways can 8 different books be distributed among 3 students so that one student gets 4 books and the other two get 2 books each?

(A) 1260
(B) 2520
(C) 630
(D) 420

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q19. A committee of 6 is to be formed from 7 men and 6 women such that the committee has more men than women. In how many ways can this be done?
(A) 658
(B) 686
(C) 700
(D) 623
Correct Answer: Option (A) : 658
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q20. Bag A has 4 red and 6 black balls; Bag B has 7 red and 3 black balls. A bag is chosen at random and one ball is drawn from it. Given that the ball drawn is red, what is the probability that it came from Bag A?
(A) 4/11
(B) 7/11
(C) 2/5
(D) 1/2
Correct Answer: Option (B) : 7/11
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q21. A drawer has 5 red pens, 4 blue pens and 3 black pens. Four pens are drawn at random. What is the probability that exactly 2 are red and 2 are blue?
(A) 4/33
(B) 3/33
(C) 5/33
(D) 4/29
Correct Answer: Option (A) : 4/33
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q22. In how many ways can 6 men and 4 women be seated at a round table such that no two women sit together?
(A) 43,200
(B) 86,400
(C) 21,600
(D) 14,400
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q23. In how many ways can the letters of the word ORANGE be arranged so that the vowels occupy only the odd positions?
(A) 36
(B) 72
(C) 18
(D) 144
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q24. Three machines A, B and C produce 25%, 35% and 40% of a factory's output respectively, with defect rates of 5%, 4% and 2%. An item is picked at random and found defective. What is the probability it came from machine C?
(A) 16/69
(B) 8/69
(C) 23/69
(D) 25/69
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q25. 5 boys and 3 girls are seated randomly in a row of 8 chairs. What is the probability that all 3 girls sit together?
(A) 3/28
(B) 1/8
(C) 3/56
(D) 5/28
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q26. A committee of 5 is to be formed from 6 men and 5 women. A specific man X must be included, a specific woman Y must be excluded, and the committee must include at least 1 woman. In how many ways can this be done?
(A) 121
(B) 126
(C) 120
(D) 115
Correct Answer: Option (A) : 121
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q27. In how many ways can the 8 distinct letters of COMPUTER be arranged such that C and T have exactly 2 letters between them (in either order)?
(A) 7200
(B) 6000
(C) 8400
(D) 5040
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q28. Two tickets are drawn one after another, without replacement, from a box containing tickets numbered 1 to 10. What is the probability that the number on the second ticket is greater than the number on the first?
(A) 1/2
(B) 9/20
(C) 11/20
(D) 2/5
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q29. A mini-deck has 16 cards: 4 suits with 4 cards each. 4 cards are drawn at random. What is the probability that at least 2 of them share the same suit?
(A) 391/455
(B) 64/455
(C) 327/455
(D) 201/455
Correct Answer: Option (A) : 391/455
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: *Mathematical Critical Thinking*

Q30. In how many ways can 8 people be seated around a circular table so that two particular people are NOT seated next to each other?

(A) 3600
(B) 5040
(C) 1440
(D) 4320

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: *Mathematical Critical Thinking*

Q31. Three fair dice are rolled. Given that at least one die shows a 6, what is the probability that the sum of the three dice is 12?

(A) 15/91
(B) 25/216
(C) 1/8
(D) 10/91

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: *Mathematical Critical Thinking*

Q32. A disease affects 1% of a population. A test detects the disease in 95% of those who have it, but also gives a false positive for 10% of those who do not. If a randomly chosen person tests positive, what is the probability that they actually have the disease?

(A) 19/217
(B) 19/100
(C) 95/1000
(D) 1/11

Correct Answer: Option (A) : 19/217

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.